

ECE 3340

Numerical Methods

Exam 3A

Name: _____

Random Numbers _____ / 20

Derivatives _____ / 20

Differential Equations _____ / 30

Numerical Integration _____ / 30

Total _____ / 100

Random Numbers. Calculate the parameters for an LCG that will provide the maximum period for a 3-bit random number generator. Show each value in the sequence starting with $x_0 = 1$. **[20 points]**

$m =$

$c =$

$a =$

Derivatives. Calculate the gradient of the explicit function $\nabla\phi(2, -1)$ represented by the sample points in the grid using 2nd order finite differences. The gradient of a function is defined as it's partial derivative along each axis: **[20 points]**

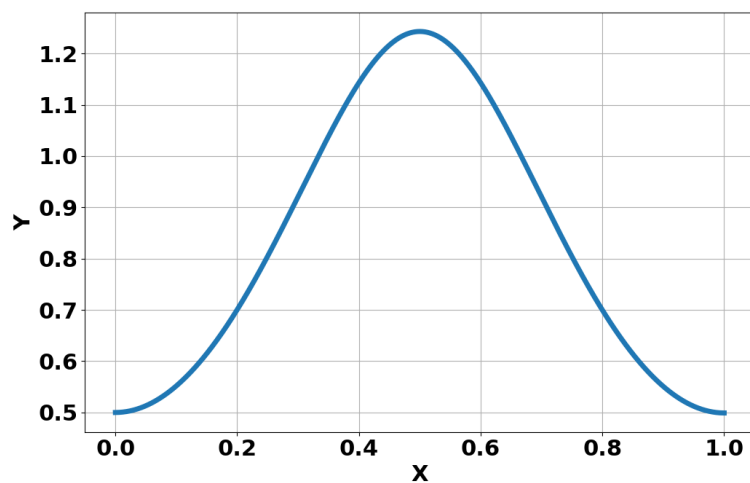
	-4	-3	-2	-1	0	1	2	3	4
-3	3.2	2.1	1.8	0.8	1.0	0.8	1.8	2.1	3.2
-2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2
-1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1
0	2.0	1.0	0.0	-1.0	-2.0	-1.0	0.0	1.0	2.0
1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1
2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2
3	3.2	2.1	1.8	0.8	1.0	0.8	1.8	2.1	3.2

$$\nabla\phi = \begin{bmatrix} d\phi/dx \\ d\phi/dy \end{bmatrix}$$

Explicit Integration. Solve the following differential equation using explicit integration

$$f(x, y) = \frac{dy}{dx} = \sin(2\pi x) e^y$$

given the initial condition $(x, y) = (0, 0.5)$ and $\Delta x = 0.2$. Plot the resulting function alongside the real result below:



Use Euler's Method (4 steps) [40 points]

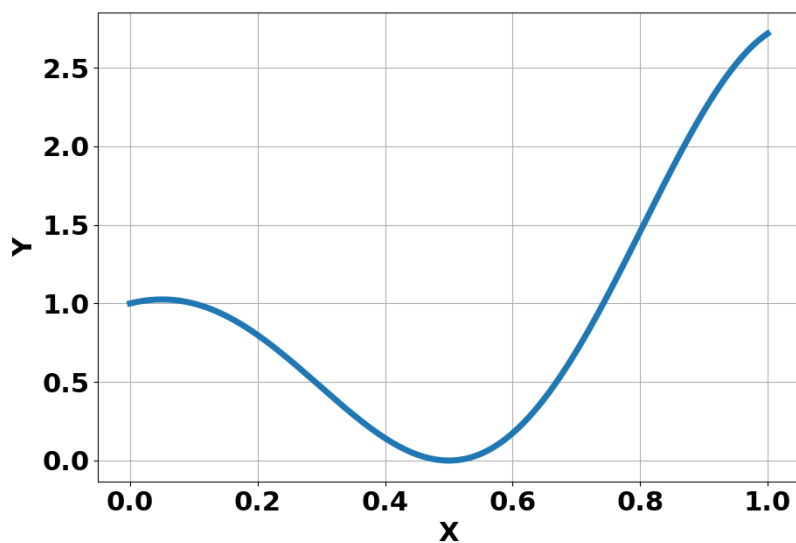
x

y

$f(x, y)$

E_r

Numerical Integration. Approximate the following integral using Simpson's rule and the Trapezoid rule. Use the same three points for the Trapezoid rule (using 2 trapezoids). **Show your work. [40 points]**



$$\int_0^1 \cos^2(\pi x) e^x dx$$